Karim ElSedfy

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Education

Bowdoin College May 2028
B.A. in Computer Science and Mathematics *Brunswick, ME*

- **GPA:** 3.85/4.00
- **Clubs and Organizations:** ColorStack, CodePath, Men's Club Rugby
- **Relevant Coursework:** Deep Learning, Artificial Intelligence, Computational Complexity Theory, Algorithms, Computer Systems, Data Structures I & II, Linear Algebra

Experience

Dropbox May 2026 – Aug. 2026
Software Engineer Intern *Remote - San Francisco, CA*

- Shipped two production growth experiences in React, TypeScript, and Python guiding Dropbox Plus users toward team subscriptions, reaching about 350K users weekly and lifting Standard-plan conversion 2.3x over the prior experience.
- Drove 10% lift in new net ARR (∼$1.4M projected annualized) by shipping the flagship growth launch four weeks ahead of schedule.
- Delivered a second production launch beyond original scope, owning architecture, full-stack implementation, instrumentation, and rollout.
- Won two of three company hackathon awards leading an AI system that turned customer reports into privacy-safe engineering investigations, correlating product activity, production errors, code changes, and issue history.

Palmera Systems Co. Jun. 2025 – Aug. 2025
Software Engineer Intern *Cairo, Egypt*

- Architected face-recognition and anti-spoofing for the smartIN app (8K daily users), leading dataset curation and model fine-tuning with DeepFace/OpenCV to reach 96% detection accuracy.
- Built responsive web frontends and backend REST services on PostgreSQL matching live captures in under 2s, cutting authentication latency by 50%.

Ungated Research, Bowdoin College Mar. 2026 – Jun. 2026
Software Engineer *Brunswick, ME*

- Co-maintained a multithreaded Python ETL pipeline (SQLAlchemy, MariaDB, AWS S3) ingesting economics papers from 11+ journals, enriched with LLM-generated summaries, tags, and translations.
- Built error-detection and automated-repair diagnostics, and ran Crossref API polling across 10 years of publications to surface 6,080 new articles for ingestion.

Computer Science Department, Bowdoin College Sep. 2025 – Dec. 2025
Independent Study Researcher *Brunswick, ME*

- Researched transferring detection-trained YOLO encoders to semantic segmentation without retraining the backbone, using dense stride-1 inference, frozen-encoder decoders, and U-Net-style skip connections.
- Improved mean IoU from 0.600 to 0.684 on the CUB-200 dataset and wrote the results as an IEEE-format paper under Prof. Jeova Farias.

Projects

Lemma Notes AI: Handwriting-Native AI Notebook GitHub | Video Demo
Swift, Vision, PencilKit, Gemini, Cloud Run, Firestore

- Built an iPad notes app where users lasso a handwritten question and receive an AI-generated answer rendered directly on the page in their own handwriting.
- Engineered the Swift AI pipeline with on-device Vision OCR, Gemini behind a Cloud Run proxy, schema-constrained responses, and deterministic ink rendering; validated the 40,000-line app/server system with 960+ automated tests.

Proteus: Virtual Try-On Frontend | Backend | Live Demo
React, Node.js, Express, Redis, PostgreSQL, Python, PyTorch, Kubernetes, KEDA, S3

- Built an event-driven, GPU-accelerated virtual try-on system where a Redis queue and pub/sub feed a Python/PyTorch worker running CatVTON on Linode Kubernetes Engine.
- Delivered real-time product UX with WebSocket completion events, presigned S3 uploads, and backend job orchestration that lets users upload a photo and receive synthesized try-on output without page reloads.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C#
AI / Data: PyTorch, OpenCV, Hugging Face Transformers, SQLAlchemy, LLM-powered data pipelines
Frameworks and Libraries: React, Node.js, Express.js, FastAPI, Flask
Technologies and Tools: PostgreSQL, Redis, Docker, Kubernetes, AWS S3, GCP, Firebase